Metrics and Evaluation Results on TradeSentinel

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Results below come from *Prototype/data/eval_results_holdout.json* (70 synthetic reports, one run on frozen settings). Training set: 108 reports. Temperature, chosen by cross-validation on training data only: 0.25. Development set result, for comparison: 95% category accuracy on 20 reports.

1. Metric Definitions

Metric	Definition	Source in pilot	Used for
Time to first correct triage	Ticket open to the first decision that experts later agree with	Ticket and audit timestamps	Value; Gate 1 and 3
Critical recall	Share of true S1/S2 issues given final severity S1/S2	Expert labels	Safety; all gates
Strict under-triage	True S1/S2 issues given any lower severity	Expert labels	Safety; all gates
False escalation rate	Share of true S3/S4 issues given S1/S2	Expert labels	Cost; alert fatigue
Calibration error (ECE)	Average gap between confidence and accuracy, over five bins	Model logs, labels	L2 eligibility
Top-3 cause hit	Final cause appears in the top three hypotheses	Closure records	Assistant value
Override rate	Decisions changed by a person, per category	Audit log	Trust; Gate 2 and 3
Median review time	Open to approve, per decision	Audit log	Rubber-stamp detection
Net payoff per issue	Share of each outcome times its payoff	Labels and finance	Gate 3

2. Category Results by Class

Category	Reports	Correct	Accuracy
order rejected	13	11	85%
order stuck	13	13	100%
execution failed	14	12	86%
fill mismatch	10	9	90%
session connectivity	10	10	100%
access howto	10	8	80%

Overall 90%; macro F1 0.899; correct in top two 95.7%. Keyword baseline: 45.7%; macro F1 0.486.

3. Calibration

Confidence bin	Reports	Mean confidence	Actual accuracy
0.2-0.4	3	0.36	0.67
0.4-0.6	9	0.52	0.56
0.6-0.8	13	0.7	0.92
0.8-1.0	45	0.94	0.98

ECE is 0.083. On this small sample the model tends to under-state its confidence, which is the safer direction. However, the two errors that passed unreviewed had 71% and 82% confidence, so confidence alone is not an adequate safety signal. This is why floors exist. The Gate 3 threshold is 0.05.

4. Severity Decisions and Payoff

Approach	Critical escalated	Missed	False escalation	USD per issue at 1% critical	at 5%	at 10%
TradeSentinel (floors + blast radius)	97.6%	1	13.8%	+50	+196	+378
Model alone, no control layer	82.9%	7	10.3%	−24	−200	−421
Keyword rules baseline	43.9%	23	10.3%	−238	−1,273	−2,567
Escalate everything	100%	0	100%	−98	+108	+365
Escalate nothing	0%	41	0%	−460	−2,462	−4,964

Formula: net payoff = p × (recall × 5,000 + (1 − recall) × (−50,000)) + (1 − p) × (FE × (−150) + (1 − FE) × 40), where p is the critical share and FE the false escalation rate. Critical recall has a 95% interval of 87.4% to 99.6% (Wilson; 40 of 41), so small differences between approaches are not meaningful.

5. Other Results

Measure	Result
Under-triaged (strict)	H49 (S1 to S2), H50 (S1 to S2), H68 (S2 to S3)
Over-triage of minor issues	17.2%
Duplicate pairs	33 correct, 0 false merges, 0 missed; 70 reports became 55 incidents
Cause ranking	Top 1: 80.6%; top 3: 96.8% of 62 reports. Misses: H09, H66
Sent to SME check	34.3% of reports. This load must be checked in the pilot
Wrong category, not sent to SME check	H09 (71% confidence), H68 (82% confidence)
Guardrail invariants	5 of 5 held on every ticket

6. Limits of This Evaluation

- One author wrote the training data, test data and incident catalogue, so shared vocabulary probably flatters results.
- The 41 critical reports are too few to support a claim of a very low miss rate.
- The critical share in the test set is artificial, so payoffs are re-weighted to assumed shares.
- The classifier stands in for a language model, whose errors may differ.
- The holdout has now been seen; fixes need a new test set.